COUNTIES REVENUE FORECASTING MODEL

Data-Driven Framework for Own-Source Revenue Forecasting

Prepared for County Governments

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Abstract

We develop a data-driven, Python-based forecasting framework for county Own-Source Revenue (OSR) that uses machine-learning diagnostics to characterize county revenue–economy structures before model selection. The evidence shows that commonly used point elasticities of OSR with respect to Gross County Product (GCP) are highly unstable**,** frequently switch sign, and largely reflect noise—confirming a practical decoupling between short-run GCP movements and OSR outcomes in many counties—while effective rates (OSR/GCP) and the OSR composition by tax head are comparatively stable. Building on these diagnostics, we estimate four simple competing models per county (trend-only, elasticity-only, trend + elasticity, and trend + growth) and select the optimal county-specific specification using **R^2 and MAPE**, retaining economic linkage only where it improves predictive performance and yields plausible parameters. Results consistently favor trend-based or constrained models over pure elasticity specifications, delivering high predictive accuracy and an operational workflow that forecasts total OSR from the selected model and allocates it across tax heads using stable observed shares—supporting credible budgeting, appraisal, and intergovernmental fiscal planning.

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1 Introduction

County governments rely on **Own Source Revenue** (OSR) to finance service delivery, support development spending, and strengthen fiscal autonomy. As counties mature institutionally, OSR has shifted from a residual financing item to a core budget pillar. This elevates the importance of credible OSR forecasting—not only for annual budget preparation, but also for program appraisal, medium-term fiscal planning, and assessment of fiscal sustainability. Weak forecasts undermine budget credibility, distort expenditure prioritization, and complicate cash and commitment management.

Effective forecasting, however, cannot be achieved by extrapolating past revenue totals alone. OSR reflects a combination of administrative capacity, revenue base composition, demographic pressure, and local economic structure. These factors evolve slowly and differ markedly across counties. A forecasting approach that ignores this structural heterogeneity risks confusing short-term fluctuations with long-run revenue capacity, and policy or administrative shifts with genuine economic fundamentals. Establishing the structural position of each county is therefore a necessary analytical first step.

To address this, the analysis adopts a data-driven structural diagnostic, using machine-learning tools. Counties are grouped using a K-means clustering algorithm, applied to standardized indicators of OSR per capita, GCP per capita, effective revenue rates (OSR/GCP), and density characteristics. The purpose is not to forecast directly, but to identify counties that exhibit similar revenue–economy structures**,** and to distinguish settings where economic linkage is strong, weak, or largely absent. The elbow plot shows a clear flattening of within-cluster variance at **k = 3**, beyond which additional clusters add little explanatory power. This choice balances statistical robustness with interpretability and produces economically meaningful groupings.

The resulting clusters in figure 1 reveal three distinct structural regimes. First, a high-capacity cluster, dominated by counties such as Nairobi, and Mombasa**,** characterized by high GCP per capita and high OSR per capita. These counties operate on a higher revenue frontier, reflecting diversified revenue bases, stronger administrative systems, and deeper monetization of local economic activity. Second, **a** middle-capacity cluster, including counties such as Nyandarua, Kirinyaga, Laikipia, Kakamega, and Bungoma, where OSR per capita is moderate but grows steadily, often supported by strong

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administrative effort rather than scale alone. Third, a low-capacity cluster, comprising counties such as Mandera, Samburu, Wajir, Marsabit, Turkana, and Busia, where both GCP and OSR per capita are low and revenue mobilization is constrained by narrow bases and structural limitations.

Crucially, while OSR per capita and GCP per capita differ sharply across clusters, the effective revenue rate (OSR/GCP) is remarkably stable within each cluster**.** Boxplots and cluster profiles show narrow dispersion in effective tax rates relative to the wide variation in economic scale. Scatter plots of log(OSR) against log(GCP) reveal parallel but distinct revenue paths: high-capacity counties lie on a higher intercept, low-capacity counties on a lower one, but counties do not switch regimes over time. This indicates that OSR performance is structurally anchored, reflecting persistent institutional and administrative characteristics rather than short-run economic movements.

These findings have direct implications for forecasting. They demonstrate that a single national elasticity or uniform GCP-based model is inappropriate**.** Counties do not share a common revenue–economy relationship, and economic growth does not translate mechanically into OSR growth across all settings. Instead, forecasting models must be conditioned on structural position: trend-based models may be sufficient in some clusters, elasticity-augmented models only where co-movement is empirically strong, and hybrid or constrained models where administrative effort dominates. In short, model choice must follow structural diagnosis, and effective OSR forecasting requires data-deep, county-specific insights grounded in how revenue systems actually function.

The rest of the analysis proceeds in a structured and sequential manner. First, we test the practical usefulness of the main approaches currently used—explicitly or implicitly—in county OSR forecasting, namely point elasticities, effective tax rates, and GCP-anchored scaling. Point elasticities *(ΔlnOSR /ΔlnGCP)* are examined and shown to be **highly unstable across years and counties**, often **switching sign** and magnitude in ways that are inconsistent with observed revenue behavior. Effective revenue rates (OSR/GCP) are then assessed as an alternative anchor. While these rates display notable stability over time, they are informative for forecasting only in counties where OSR and GCP move together persistently. The analysis further demonstrates that mechanically anchoring OSR to GCP growth is not universally valid: in several counties, revenue growth is more closely linked to administrative effort, enforcement, and base management than to measured economic output, suggesting that GCP alone does not fully capture taxable activity at the county level.

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Building on these diagnostics, the analysis shifts from a uniform modelling approach to county-specific optimal model selection, guided by empirical performance. Four candidate model families are estimated for each county, reflecting different revenue-generation mechanisms, ranging from pure trend models to carefully constrained economic-linkage specifications. Models are evaluated using a transparent accuracy framework that priorities forecast performance **(MAPE)** while ensuring adequate explanatory power (**R^2**). This results in differentiated model choices across counties: trend-dominant counties are forecast using structurally stable trend models, while counties with proven OSR–GCP co-movement are assigned models that incorporate economic linkage. The outcomes are summarized in Tables 1–3, culminating in OSR forecasts for FY2025 and FY2026 that are empirically grounded, operationally practical, and aligned with county-specific revenue structures.

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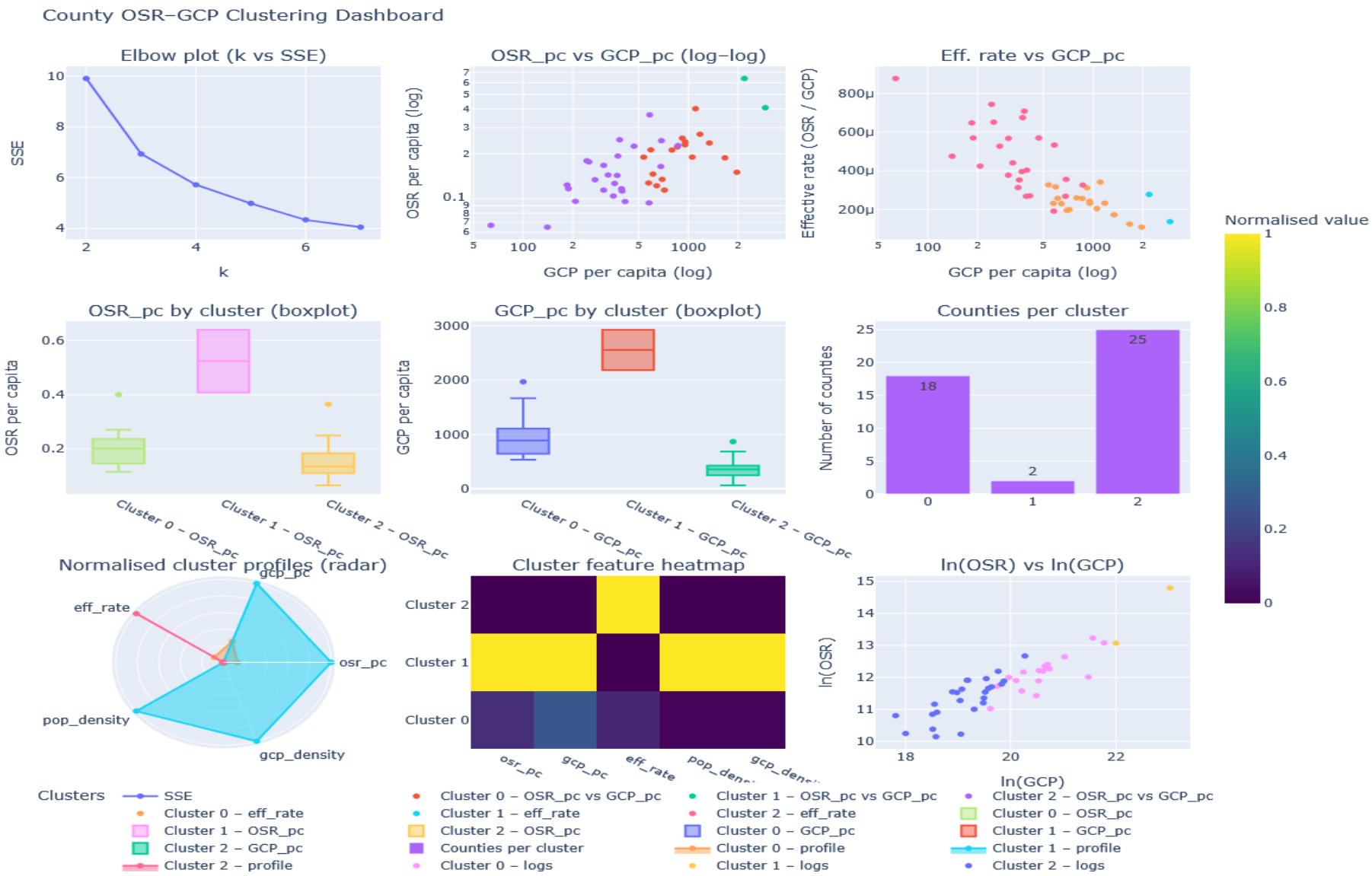
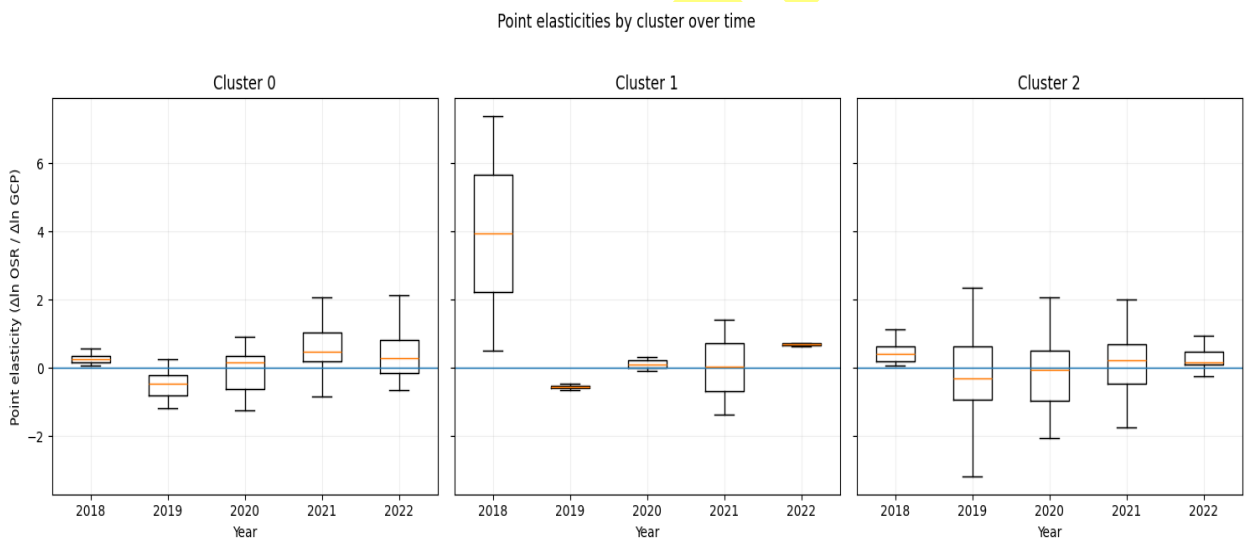
Figure 1Dashboard 1



2. Re-examining the Use of Point Elasticities in County OSR Forecasting

Preliminary review of county revenue forecasting practices shows that point elasticities of OSR with respect to GCP are widely used as the primary mechanism for linking economic activity to own-source revenue projections. In principle, this approach assumes that changes in county economic output translate proportionally and predictably into revenue growth. However, the evidence presented in *Figures 2 &3* demonstrates that this assumption does not hold in practice. Across all three structural clusters, median elasticities fluctuate sharply from year to year, with interquartile ranges widening rather than narrowing over time. For example, at the cluster level (*Figure 2*), median elasticities shift from positive to negative between consecutive years, while dispersion frequently exceeds ±1, indicating that elasticity estimates are not stable even within broadly similar county groups.

Figure 2 point elasticities box plots by clusters



At the county level, the instability is more pronounced and often extreme (*Figure 4*). Counties such as Elgeyo-Marakwet, Kisii, and Nandi record elasticity spikes exceeding +20 in individual years, followed by negative or near-zero values in subsequent periods. In Turkana and Wajir, elasticities swing sharply despite relatively smooth revenue paths, reflecting sensitivity to short-term GCP movements rather than genuine changes in revenue capacity. Conversely, counties like Meru, Nyandarua, and Siaya exhibit elasticities clustered tightly around zero even as OSR grows steadily year after year. These patterns show that elasticity estimates are dominated by denominator volatility and timing effects, not by structural revenue–economy relationships.

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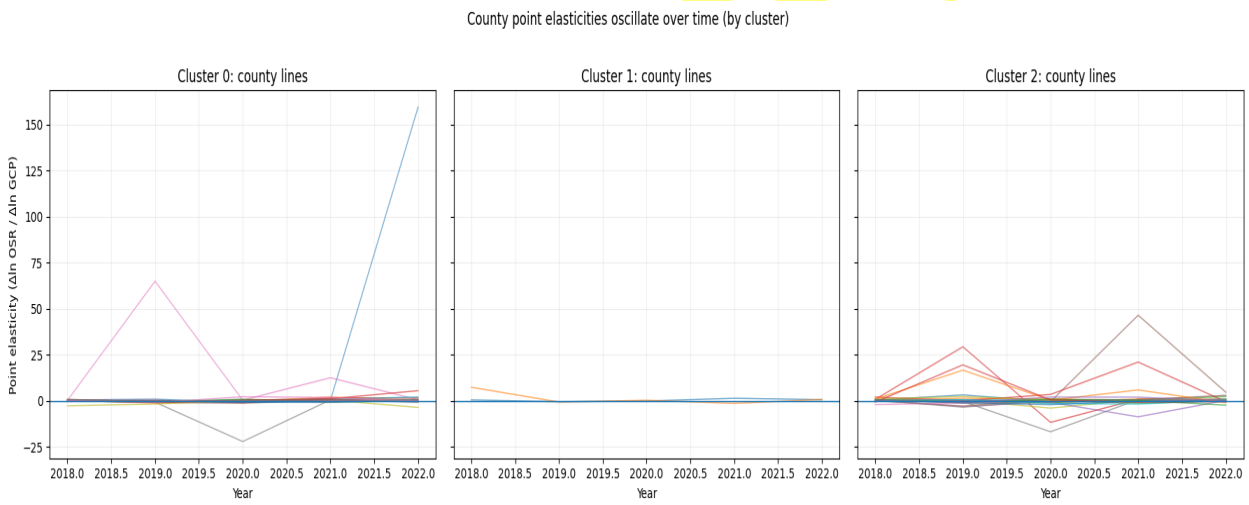
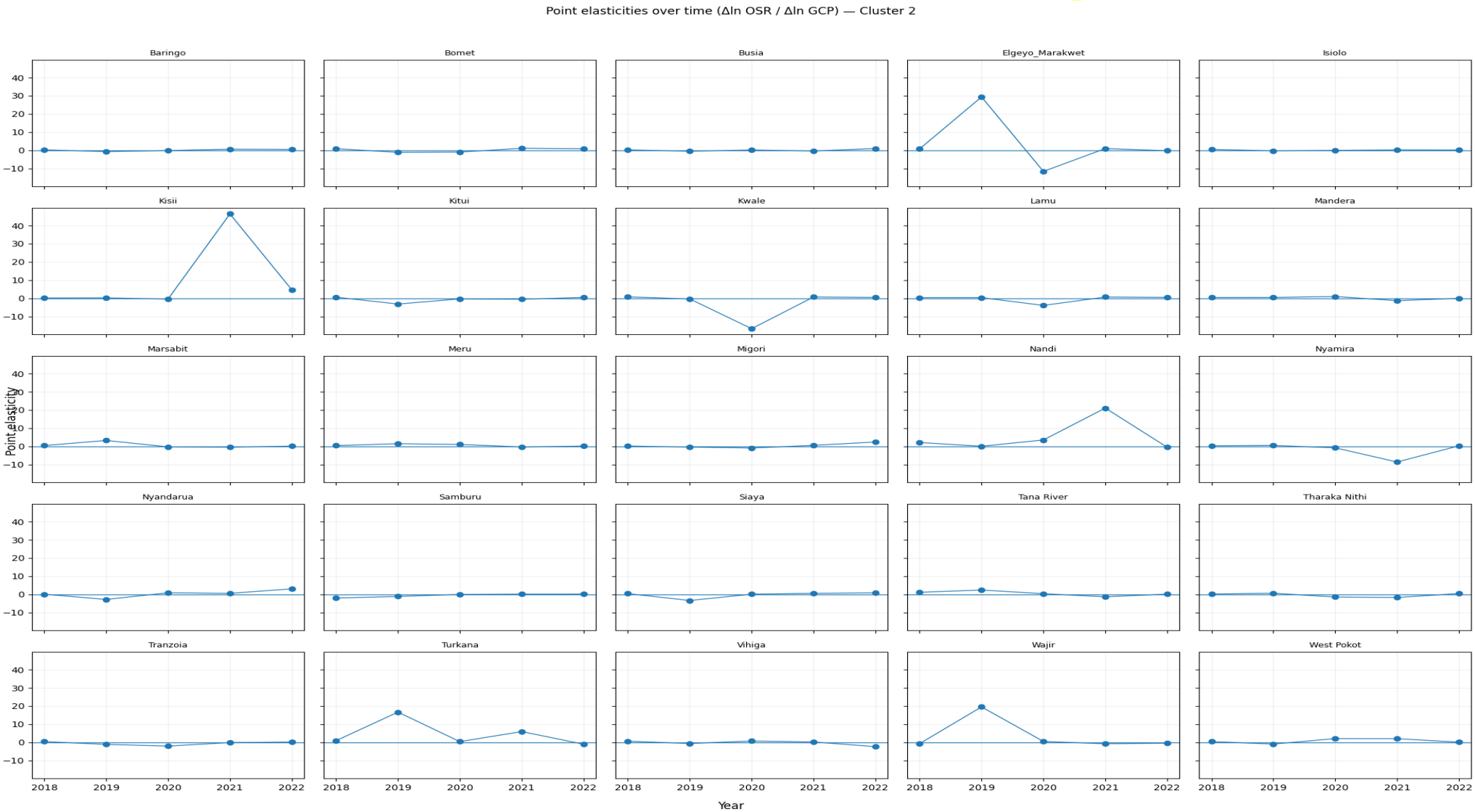
Further, the point-elasticity results in Figure 3 classified by cluster and figure 4, county specific trends underscore why reliance on year-to-year elasticities is problematic for OSR forecasting. Across all three clusters, elasticities exhibit sharp spikes, collapses, and frequent sign reversals. In Cluster 0 and Cluster 2 in particular, elasticities swing from strongly positive to negative within short periods, with some counties registering extreme values that are clearly disconnected from underlying revenue trends. Negative elasticities—where OSR falls despite rising GCP, or rises when GCP contracts—signal decoupling between economic output and revenue collection, rather than genuine behavioral responses. Even in Cluster 1, where co-movement is relatively stronger, elasticities remain unstable and sensitive to short-term fluctuations. These patterns further confirm that point elasticities are noisy, non-robust, and unsuitable as a primary forecasting tool, reinforcing the need for structurally grounded and county-specific models instead.

Figure 3 Elasticities over clusters



Crucially, this volatility contrasts sharply with the underlying stability of OSR itself. As shown in the raw OSR trend plots (*Figure 6*), nearly all counties display smooth, monotonic revenue growth over the sample period, with no corresponding volatility in actual collections. The implication is clear: point elasticities introduce noise that is absent in the revenue data they are meant to explain. As a result, forecasts based on these elasticities risk over-reacting to temporary economic fluctuations and systematically misrepresenting revenue potential. This empirical failure of point elasticities motivates the need for a more robust modelling framework—one that anchors forecasts in stable trend behavior and incorporates economic linkage only where it is demonstrably persistent and improves predictive performance.

Figure 4 Elasticities by county



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3. Effective Tax Rates as a Conditional Anchor for OSR Forecasting

Given the instability of point elasticities, the analysis next turns to effective OSR/GCP rates as a potential alternative anchor for forecasting. Unlike elasticities, effective tax rates capture the average revenue extraction intensity embedded in county systems—reflecting tax bases, fee structures, enforcement effort, and administrative capacity. Figure 5 shows that, at an aggregate level, these rates are markedly more stable over time. Cluster-level averages evolve smoothly with no abrupt reversals: Cluster 1 consistently exhibits the highest extraction intensity (rising from approximately 0.00045 to above 0.00060), Cluster 0 remains anchored around 0.00020–0.00030, and Cluster 2 operates at a lower but steady range of roughly 0.00013–0.00022. The cluster boxplots reinforce this finding: within-cluster medians shift gradually, and dispersion remains contained relative to the extreme volatility observed in point elasticities.

At the county level, effective rates also display temporal anchoring. Counties such as Bomet, Kisii, Kirinyaga, and Elgeyo-Marakwet show gently rising or stable effective rates alongside steady revenue growth, while Kitui, Kwale, and Embu maintain almost flat extraction profiles over the entire period. Even in counties with visible movement—such as Garissa or Marsabit—changes are incremental rather than erratic. This behaviour confirms that OSR performance is governed primarily by structural and institutional factors, not by short-run economic shocks. In this sense, effective tax rates are empirically superior to elasticities as descriptive measures of revenue systems.

However, stability alone does not make effective tax rates universally useful for forecasting. The figures also make clear that in counties where OSR and GCP are decoupled, effective rates lose their economic meaning. In Nairobi, Wajir, and Samburu, for example, effective rates remain stable even as GCP movements bear little or no relationship to OSR outcomes. In such cases, multiplying projected GCP by a stable effective rate produces mechanically consistent but economically empty forecasts. Effective tax rates are therefore only informative when OSR and GCP demonstrably co-move. This insight directly informs the modelling strategy adopted next: effective-rate–based models are used selectively, only where economic linkage is empirically established; elsewhere, forecasting must rely on trend-based specifications that reflect institutional revenue capacity rather than assumed economic responsiveness.

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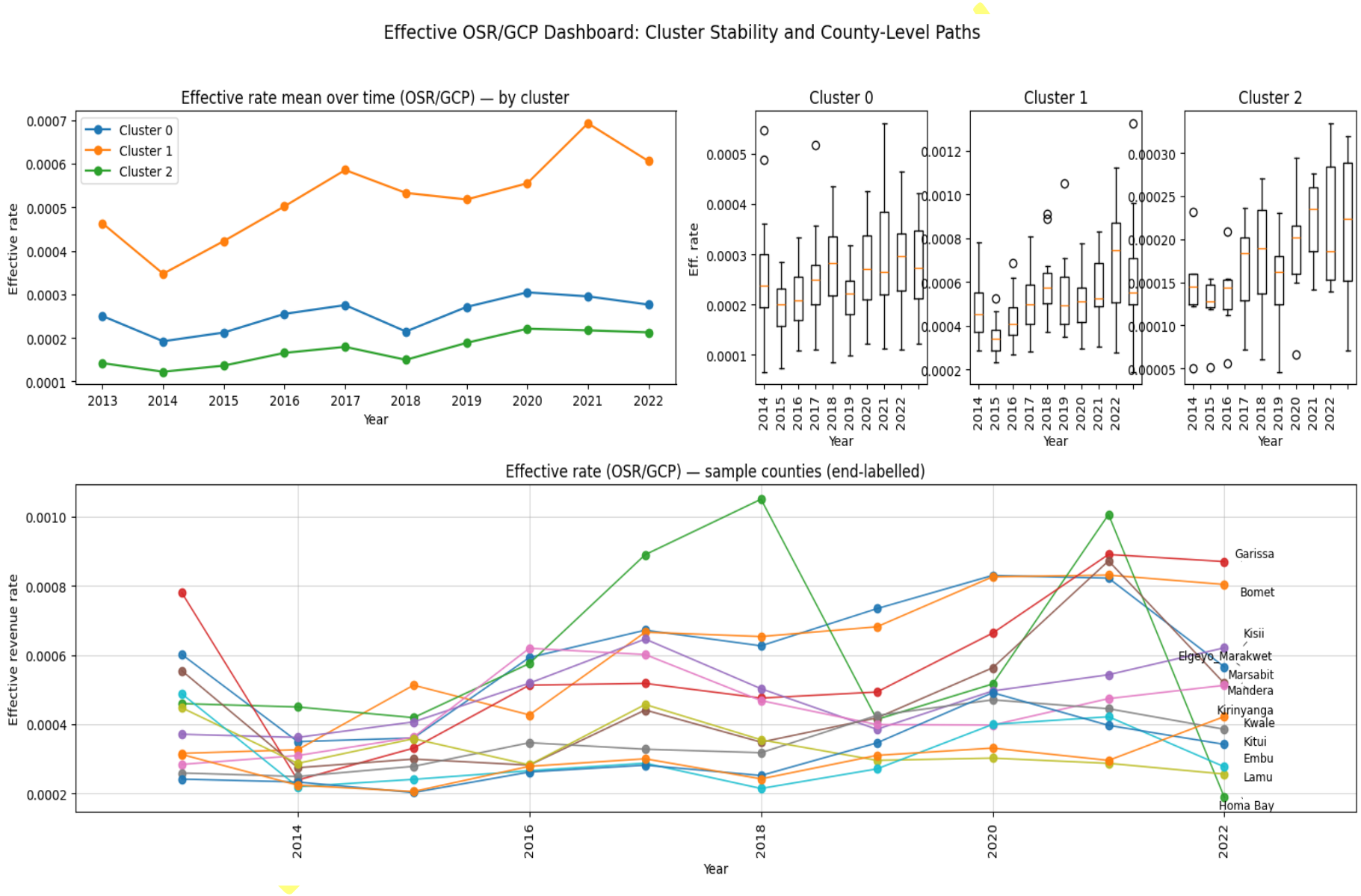
Figure 5 effective tax rates



4. Assessing Co-movement and Structural Alignment

To formally assess whether economic activity can meaningfully anchor OSR forecasting, both OSR and GCP are transformed using county-specific min–max scaling to a common 0–1 range and plotted jointly over time (Figure 6). This transformation preserves each series' internal dynamics while removing level effects, allowing direct comparison of direction, timing, and co-movement rather than magnitude. The objective is not to compare counties to one another, but to evaluate whether, within a county, revenue and economic activity evolve together in a consistent and economically interpretable way.

The results reveal clear structural segmentation. In a subset of counties—including Bomet, Kirinyaga, Kisii, Bungoma, Kakamega, and Nyeri—scaled OSR and GCP move closely together. Turning points broadly align, and revenue responds in the same direction as economic activity. In these cases, economic growth carries real informational content for revenue forecasting, and GCP can legitimately enter forecasting models—either through elasticity-based terms or growth-adjusted extensions layered onto a trend. For these counties, effective tax rates retain economic meaning because they operate on a base that is demonstrably linked to observed revenue outcomes.

In contrast, many counties exhibit weak, intermittent, or outright reversed co-movement. Counties such as Nairobi, Wajir, Samburu, Kwale, Kitui, Marsabit, and Mandera show prolonged periods where GCP rises while OSR remains flat or grows smoothly, or where OSR increases despite volatile or stagnant GCP. Nairobi is a particularly instructive case: scaled OSR follows a near-monotonic upward path, while scaled GCP fluctuates sharply, indicating that revenue performance is governed primarily by administrative capacity, enforcement intensity, and tax base design rather than contemporaneous economic output. In Wajir and Samburu, divergence is even more pronounced, with no stable transmission mechanism from economic activity to revenue.

These patterns have direct implications for model choice. Even when effective tax rates are stable, they become economically uninformative if the underlying base (GCP) is decoupled from revenue behavior**.** Stability alone is therefore insufficient: without co-movement, effective rates cannot anchor forecasts in any meaningful way. What remains consistent across virtually all counties, however, is the underlying OSR trend, reflecting institutional maturity and sustained revenue mobilisation effort. This evidence motivates the modelling framework adopted next: GCP-based components are used selectively and conditionally, while trend-anchored models form the default backbone of OSR forecasting**,** ensuring robustness, transparency, and credibility in county budgeting and medium-term fiscal planning.

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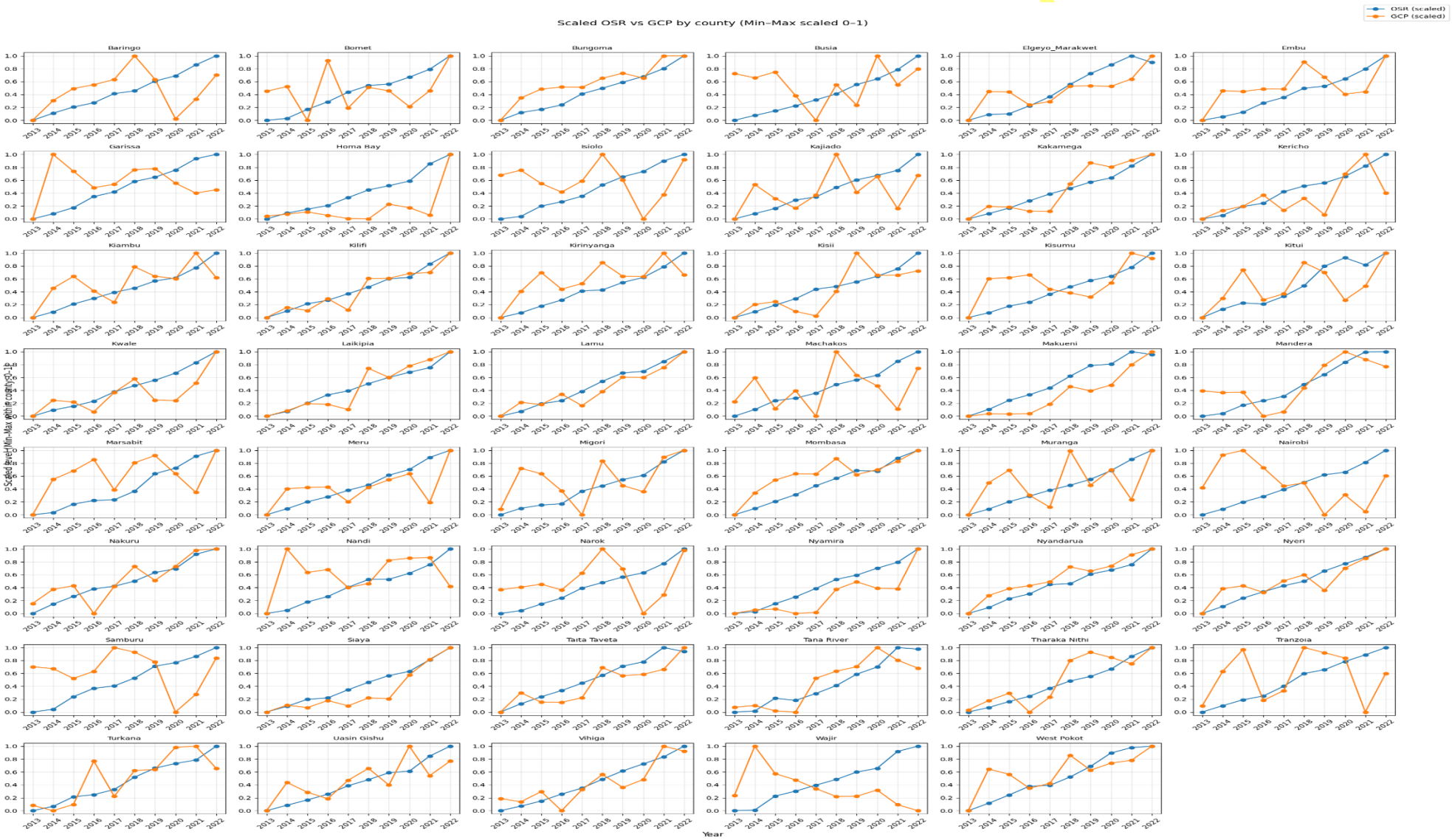
Figure 6 Revenue vs GCP trends



5 County OSR Trends, Economic Linkages, and Forecasting Implications

Figure 7 shows raw Own Source Revenue (OSR) trajectories for all counties without any scaling or transformation. Across the board, OSR follows a smooth, upward path, with very limited year-to-year reversals. Counties such as Bungoma, Kakamega, Kisii, Nyeri, Nyandarua, Uasin Gishu, and Bomet exhibit almost linear growth over the entire period, while even structurally larger and more complex economies—Nairobi, Mombasa, Kiambu, Nakuru—display strong persistence despite macroeconomic shocks. This visual evidence confirms that OSR is not a volatile series driven by short-run fluctuations, but a slow-moving, structurally anchored revenue process.

Second, this stability contrasts sharply with the behaviour of economic activity. When these trends are read alongside earlier figures comparing OSR and GCP (scaled and unscaled), it becomes clear that OSR growth is often only weakly synchronized with changes in GCP. In several counties—such as Machakos, Kajiado, Kitui, Marsabit, Samburu, and Wajir—GCP exhibits noticeable oscillations, while OSR continues to rise steadily. Even in counties where co-movement exists (for example Nyandarua, Bungoma, or Kakamega), the relationship is gradual rather than contemporaneous. This gap between smooth revenue growth and more volatile economic activity explains why point elasticities computed year-by-year appear unstable and frequently counterintuitive.

Third, the weak short-run linkage has direct implications for forecasting practice**.** Forecasting frameworks that rely mechanically on GCP-based elasticities implicitly assume that revenue responds quickly and proportionally to economic changes. Figure 7 shows that this assumption does not hold for most counties. Where OSR continues to rise even during periods of muted or declining GCP, elasticity-based forecasts will systematically under-predict revenue. Conversely, in years of strong economic expansion, they risk over-prediction. This is not a modelling error; it reflects the institutional reality that county revenues are driven by administrative capacity, compliance systems, tariff structures, and enforcement—factors that evolve slowly over time.

The evidence points clearly to trend-based models as the appropriate baseline for county OSR projections. Economic variables such as GCP can play a supporting role, but only where diagnostics confirm a stable and meaningful long-run relationship. Treating GCP as a universal driver of OSR is not empirically defensible. Instead, forecasting should begin with the strong deterministic trend observed across counties, and then selectively incorporate economic linkages where they improve accuracy. This logic underpins the econometric model selection that follows, where alternative specifications are tested systematically and the optimal forecasting model is chosen county by county.

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Figure 7 trends of revenue

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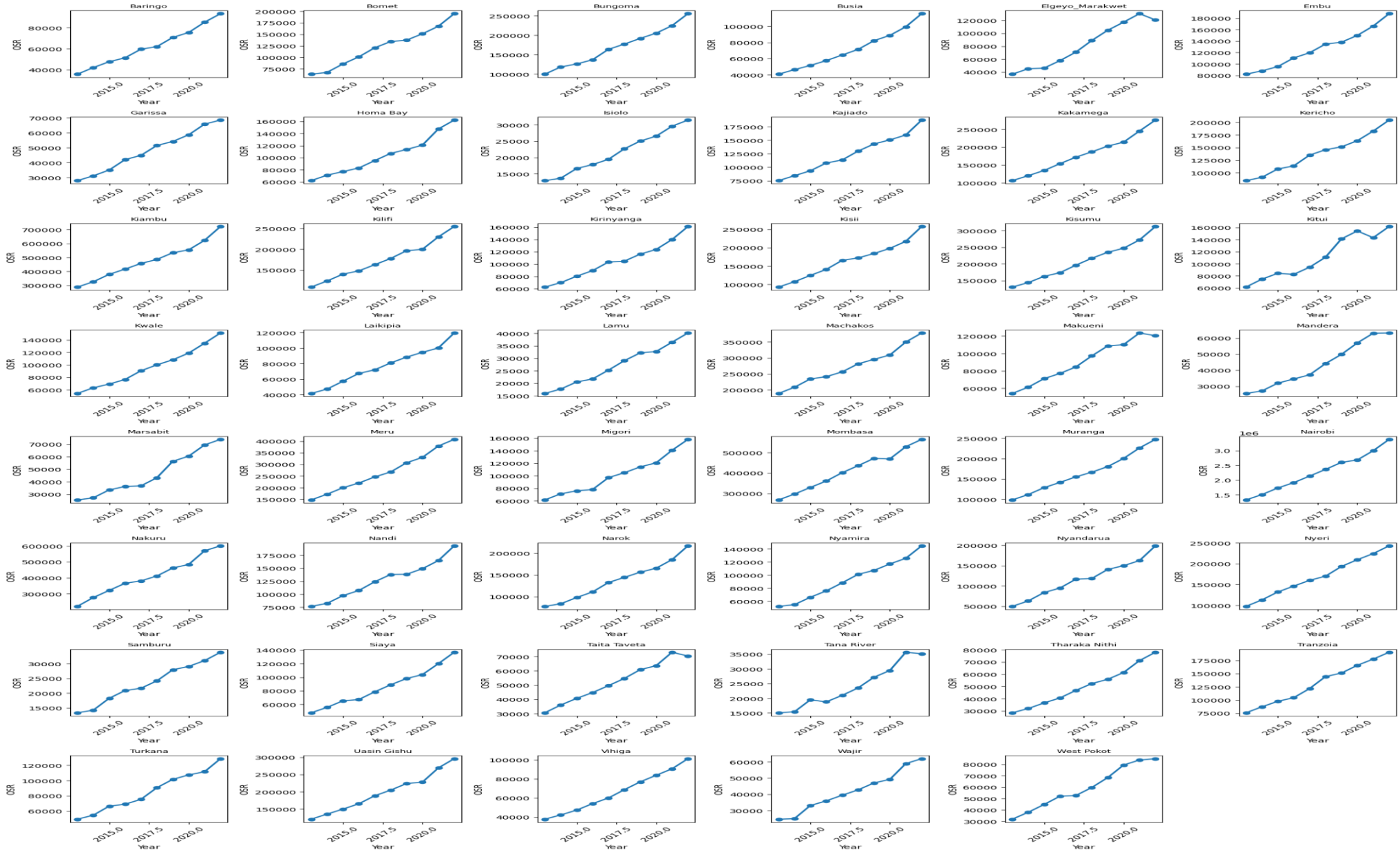
Raw OSR Trends by County (No Scaling)



6. Correlation and Effective Tax Rates: Why Co-movement Is Not Automatic

Figure 8 exposes a decisive structural break: correlation strength and effective tax rates do not move in tandem**.** The horizontal bars report the correlation between ln(OSR) and ln(GCP), while the dots represent average effective tax rates (OSR/GCP). If OSR were fundamentally driven by economic activity, counties with strong positive correlations would also display systematically higher—or rising—effective tax rates. The data reject this hypothesis. Counties such as Nyandarua, Kilifi, Laikipia, Kakamega, and Bungoma show relatively high correlations, yet their effective tax rates are not materially different from counties with weak or unstable correlations. Conversely, counties with negative correlations—notably Nairobi, Wajir, Samburu, and Isiolo—continue to post steady and rising OSR, with effective tax rates that remain non-trivial and broadly stable. This immediately breaks the presumed causal chain from GCP → OSR → effective tax rate.

The presence of negative correlations alongside sustained revenue growth is the critical diagnostic result. These are not collapsing or fiscally distressed counties. Rather, they are counties where OSR growth **is** administratively driven and structurally anchored, evolving independently of short-run economic fluctuations captured by GCP. In such cases, both correlation and effective tax rates lose their legitimate economic interpretation: correlation reflects noise, while effective rates—though numerically stable—no longer represent extraction from a meaningful economic base. The clustering of effective tax rates across counties with widely different correlation coefficients further confirms that revenue intensity is governed by institutional design, enforcement effort, and revenue base composition, not contemporaneous output movements.

From a forecasting standpoint, this figure draws a hard methodological boundary. GCP can only be used as a conditioning variable where correlation is both positive and sufficiently strong—operationally, where the ln(OSR)–ln(GCP) correlation is consistently above 0.5. Outside this narrow set of counties, elasticity-based or GCP-anchored models are not just weak—they are conceptually invalid and risk embedding noise directly into forecasts. For the majority of counties, including high-performing ones such as Nairobi**,** trend-based models dominate, with GCP playing at most a secondary, diagnostic role. This finding justifies the modelling strategy adopted next: economic variables are applied selectively, while trend dynamics form the core forecasting engine, ensuring stability, credibility, and fiscal realism.

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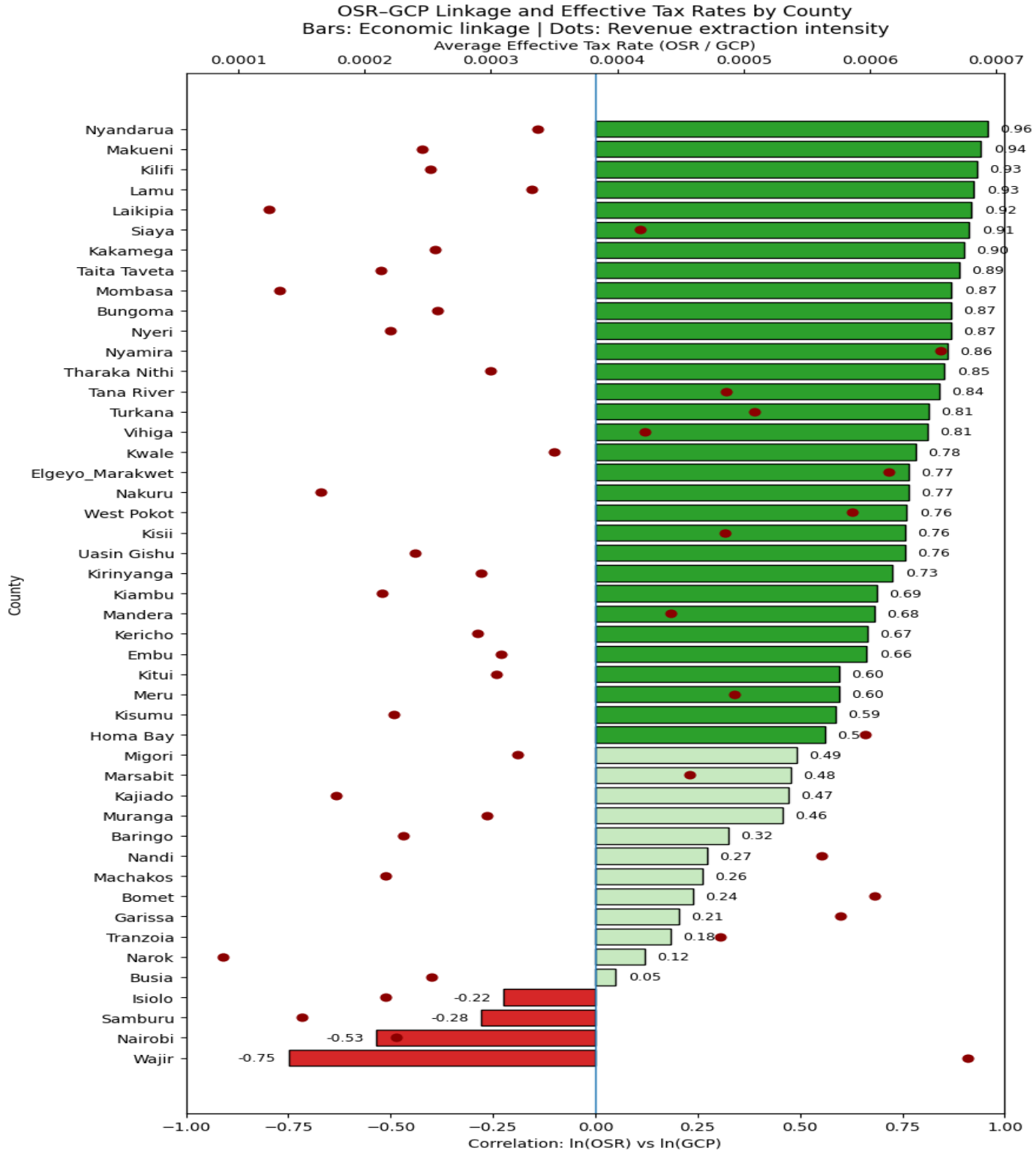
Figure 8 OSR and GCP correlations



7. Identification of Optimal County-Level Forecasting Models

Building on the structural diagnostics, the forecasting stage adopts a county-specific model selection strategy rather than imposing a single rule across all counties. Four simple log-linear specifications are estimated for every county: **(A)** a *trend-only* model capturing institutional/administrative deepening; **(B)** a *pure elasticity* model linking revenue to the level of county output; **(C)** a *trend-plus-elasticity* model allowing both long-run administrative drift and economic scale to matter; and **(D)** a *trend-plus-growth* model that conditions revenue on GCP growth rather than the level. In line with the earlier evidence on decoupling, model (B) is only considered where the economic linkage is demonstrably strong—operationally, where the correlation between ln(OSR) and ln(GCP) is **≥ 0.5**—so that "economic anchoring" is not forced onto counties where it is empirically absent. The objective is not complexity, but fit-for-purpose forecasting**:** a model set that is transparent, stable under short samples (10 years), and implementable by county teams.

A. $\text{Ln (OSR)} = \alpha + \beta\cdot(t)$ 1
B. $\ln (\text{OSR}) = \alpha + \gamma\cdot \ln (\text{GCP})$. ...2
C. $\ln (\text{OSR}) = \alpha + \beta\cdot t + \gamma\cdot \ln (\text{GCP})$...3
D. $\ln (\text{OSR}) = \alpha + \beta\cdot t + \gamma\cdot \Delta \ln (\text{GCP})$ ...4

The rationale for this modelling set is empirical rather than theoretical. Earlier sections show that OSR growth is generally smooth and persistent, while point elasticities are unstable and often misleading. At the same time, effective tax rates display considerable stability within counties, suggesting that economic linkage—where present—operates through gradual, long-run channels rather than volatile year-to-year responses. The four-model framework therefore allows the **data to determine** whether revenue is best explained by time trends alone, by economic scale, by a combination of both, or by economic momentum. Importantly, all models are deliberately simple and can be implemented directly in Excel, ensuring usability by county finance teams.

Model performance is evaluated systematically and transparently. Each model is estimated separately for every county and assessed using two criteria: goodness of fit (R^2) and forecast accuracy (MAPE). A model is only retained as optimal if it improves accuracy relative to the trend-only benchmark and yields economically sensible parameters. Elasticity-based models are rejected where coefficients are unstable, implausible, or worsen forecast performance. This disciplined selection process ensures that economic variables are incorporated only where they add measurable predictive value, rather than by assumption.

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Table 1 summarizes the recommended model per county alongside estimated parameters (intercepts, trend coefficients, and—where applicable—elasticity/growth terms). Table 2 provides the decision metrics: **R^2** (explanatory power) and MAPE (forecast accuracy). The results show that county OSR is, in most cases, best explained by trend dynamics, not by contemporaneous economic output. Trend-only models dominate the recommendations, including for counties where GCP linkage is weak or negative—e.g., Nairobi (corr ≈ −0.535), Wajir (corr ≈ −0.750), Isiolo (corr ≈ −0.225), and Samburu (corr ≈ −0.279)—yet these counties still achieve high R^2 and acceptable MAPE under the trend specification, confirming that revenue growth is being driven primarily by structural and administrative factors. Across the sample, the selected models typically deliver very strong fit (R^2 commonly around the high 0.9s) and low forecast error (MAPE largely in the low single- to low double-digit range). Where economic variables do add measurable value, the gain is disciplined and modest: counties such as Kwale (recommended **C**, MAPE ≈ 0.012; R^2 ≈ 0.997), Mombasa (recommended **C**, MAPE ≈ 0.020; R^2 ≈ 0.989), Embu (recommended **C**, MAPE ≈ 0.021; R^2 ≈ 0.992), Bungoma (recommended **C**, MAPE ≈ 0.022; R^2 ≈ 0.992), Kirinyaga (recommended **C**, MAPE ≈ 0.027; R^2 ≈ 0.987), and Mandera (recommended **C**, MAPE ≈ 0.025; R^2 ≈ 0.991) show improved accuracy when GCP is introduced alongside trend. A separate group performs better with growth-conditioning (D), consistent with gradual linkage through momentum rather than levels—e.g., Tharaka Nithi (recommended D, MAPE ≈ 0.016; R^2 ≈ 0.996), Kakamega (recommended D, MAPE ≈ 0.020; R^2 ≈ 0.992), Turkana (recommended D, MAPE ≈ 0.032; R^2 ≈ 0.980), Makueni (recommended D, MAPE ≈ 0.034; R^2 ≈ 0.960), and Laikipia (recommended D, MAPE ≈ 0.036; R^2 ≈ 0.974). The common thread is that economic variables are only retained when they reduce MAPE relative to the trend benchmark and remain economically sensible—otherwise, they are treated as noise and excluded.

Table 3 then converts the chosen county-specific specifications into operational forecasts for 2025 and 2026. The framework is deliberately Excel-ready: once a county's recommended model is known (Table 1), forecasts are generated using only the estimated parameters plus (where relevant) projected GCP levels or growth. For trend-only counties, forecasts require only the year index (e.g., 2025 and 2026 mapped to the appropriate *t* values). For trend-plus-elasticity counties, forecasts require *t* and ln(GCP**)** projections. For trend-plus-growth counties, forecasts require *t* and Δln(GCP) projections. This produces a single, disciplined forecasting workflow: (i) pick the recommended model from Table 1, (ii) confirm its credibility using Table 2 (MAPE and R^2), and (iii) generate the 2025–2026 projections in Table 3 using simple Excel formulas. The outcome is a forecasting system that is empirically grounded, county-specific, transparent, and implementable—and that aligns directly with the core diagnostic conclusion of the study: GCP-based modelling is selective; trend-based structure is the default anchor.

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Table 1 model comparison per county

county	n_obs	year _min	year _max	corr_ln_ osr_ln_gcp	A_ trend_a	A_ trend_b	A_r2	A_ mape	B_ gamma	B_r2	B_ mape	C_ gamma	C_r2	C_ mape	D_ gamma_g	D_r2	D_ mape	recommended _model	best _mape
Busia	10	2013	2022	0.05	10.623	0.113	0.998	0.01										A_trend_only	
Kwale	10	2013	2022	0.783	10.926	0.112	0.996	0.017	1.444	0.613	0.177	0.114	0.997	0.012	0.075	0.997	0.014	C_trend_plus_elasticity	0.012
Tharaka Nithi	10	2013	2022	0.853	10.278	0.111	0.996	0.017	0.576	0.728	0.153				0.028	0.996	0.016	D_trend_plus_growth_gamma	0.016
Machakos	10	2013	2022	0.261	12.172	0.073	0.988	0.018										A_trend_only	0.018
Vihiga	10	2013	2022	0.812	10.552	0.111	0.996	0.018	0.943	0.659	0.16							A_trend_only	0.018
Uasin Gishu	10	2013	2022	0.756	11.723	0.097	0.991	0.018	1.144	0.571	0.156							A_trend_only	0.018
Kilifi	10	2013	2022	0.932	11.634	0.089	0.991	0.018	0.947	0.868	0.076							A_trend_only	0.018
Muranga	10	2013	2022	0.458	11.533	0.099	0.994	0.019										A_trend_only	0.019
Kisumu	10	2013	2022	0.587	11.793	0.094	0.994	0.019	0.889	0.344	0.198							A_trend_only	0.019
Kakamega	10	2013	2022	0.9	11.604	0.102	0.992	0.021	0.538	0.81	0.11				0.001	0.992	0.02	D_trend_plus_growth_gamma	0.02
Mombasa	10	2013	2022	0.87	12.538	0.081	0.982	0.028	0.894	0.757	0.111	0.152	0.989	0.02				C_trend_plus_elasticity	0.02
Baringo	10	2013	2022	0.325	10.539	0.102	0.992	0.021										A_trend_only	0.021
Meru	10	2013	2022	0.595	11.948	0.111	0.994	0.021	0.833	0.354	0.203	0.009	0.994	0.022				A_trend_only	0.021
Embu	10	2013	2022	0.663	11.312	0.091	0.991	0.022	0.484	0.44	0.161	0.033	0.992	0.021	0.025	0.989	0.022	C_trend_plus_elasticity	0.021
Bungoma	10	2013	2022	0.869	11.555	0.1	0.99	0.025	0.511	0.756	0.12	0.054	0.992	0.022	0.003	0.989	0.022	C_trend_plus_elasticity	0.022
Kajiado	10	2013	2022	0.471	11.255	0.097	0.99	0.023										A_trend_only	0.023
Homa Bay	10	2013	2022	0.562	11.041	0.104	0.991	0.023	0.285	0.315	0.209				0.007	0.988	0.025	A_trend_only	0.023
Siaya	10	2013	2022	0.914	10.809	0.112	0.992	0.024	0.567	0.835	0.11							A_trend_only	0.024
Isiolo	10	2013	2022	-0.225	9.474	0.103	0.989	0.025										A_trend_only	0.025
Mandera	10	2013	2022	0.681	10.131	0.11	0.987	0.033	0.721	0.464	0.206	0.088	0.991	0.025	0.044	0.984	0.032	C_trend_plus_elasticity	0.025
Migori	10	2013	2022	0.492	11.032	0.102	0.987	0.026										A_trend_only	0.026
Kirinyanga	10	2013	2022	0.727	11.076	0.099	0.987	0.027	0.934	0.528	0.156	0.035	0.987	0.027				C_trend_plus_elasticity	0.027
Nairobi	10	2013	2022	-0.535	14.145	0.099	0.99	0.027										A_trend_only	0.027
Kericho	10	2013	2022	0.665	11.364	0.096	0.986	0.028	1.004	0.443	0.178							A_trend_only	0.028
Kiambu	10	2013	2022	0.688	12.62	0.094	0.985	0.03	0.751	0.473	0.158	0.026	0.985	0.029				C_trend_plus_elasticity	0.029
Narok	10	2013	2022	0.121	11.267	0.113	0.987	0.029										A_trend_only	0.029
Tranzoia	10	2013	2022	0.184	11.274	0.104	0.984	0.03										A_trend_only	0.03
Nyeri	10	2013	2022	0.868	11.555	0.098	0.985	0.03	1.012	0.754	0.105	0.055	0.985	0.031				A_trend_only	0.03
Turkana	10	2013	2022	0.815	10.836	0.105	0.983	0.036	1.448	0.664	0.131	0.021	0.983	0.035	0.084	0.98	0.032	D_trend_plus_growth_gamma	0.032
Lamu	10	2013	2022	0.925	9.702	0.103	0.987	0.032	0.646	0.856	0.083							A_trend_only	0.032
Garissa	10	2013	2022	0.205	10.28	0.102	0.982	0.033										A_trend_only	0.033
Makueni	10	2013	2022	0.942	10.96	0.094	0.966	0.04	0.478	0.888	0.081				0.008	0.96	0.034	D_trend_plus_growth_gamma	0.034
Nyamira	10	2013	2022	0.861	10.871	0.115	0.983	0.034	0.8	0.741	0.148				0.051	0.978	0.037	A_trend_only	0.034
Laikipia	10	2013	2022	0.919	10.704	0.11	0.976	0.044	0.795	0.845	0.101				0.004	0.974	0.036	D_trend_plus_growth_gamma	0.036
Taita Taveta	10	2013	2022	0.888	10.402	0.095	0.973	0.037	0.68	0.789	0.108							A_trend_only	0.037
Nandi	10	2013	2022	0.273	11.27	0.098	0.978	0.037										A_trend_only	0.037
Kisii	10	2013	2022	0.756	11.503	0.105	0.977	0.038	1.074	0.572	0.178							A_trend_only	0.038
Nakuru	10	2013	2022	0.765	12.42	0.102	0.97	0.039	0.993	0.585	0.162							A_trend_only	0.039
Wajir	10	2013	2022	-0.75	10.126	0.104	0.973	0.041										A_trend_only	0.041
West Pokot	10	2013	2022	0.761	10.447	0.11	0.973	0.042	1.155	0.579	0.182	0.065	0.974	0.043				A_trend_only	0.042
Tana River	10	2013	2022	0.841	9.588	0.102	0.973	0.042	0.609	0.707	0.149							A_trend_only	0.042
Marsabit	10	2013	2022	0.478	10.116	0.125	0.979	0.045										A_trend_only	0.045

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Samburu	10	2013	2022	-0.279	9.549	0.104	0.967	0.048										A_trend_only	0.048
Nyandarua	10	2013	2022	0.958	10.967	0.14	0.958	0.072	1.088	0.918	0.105	0.409	0.977	0.05				C_trend_plus_elasticity	0.05
Bomet	10	2013	2022	0.238	11.098	0.123	0.968	0.054										A_trend_only	0.054
Kitui	10	2013	2022	0.595	11.073	0.109	0.948	0.064	1.065	0.354	0.218	0.074	0.95	0.063				C_trend_plus_elasticity	0.063
E_Marakwet	10	2013	2022	0.765	10.543	0.151	0.958	0.074	1.069	0.586	0.242							A_trend_only	0.074

Table 2 optimal model choice per county

County	Optimal Model	a (Intercept)	b (Trend)	γ (Elasticity / Growth)	R^2	MAPE
Baringo	Trend only (A)	10.539	0.102	—	0.992	0.021
Bomet	Trend only (A)	11.098	0.123	—	0.968	0.054
Bungoma	Trend + Elasticity (C)	11.555	0.100	0.054	0.992	0.022
Busia	Trend only (A)	10.623	0.113	—	0.998	0.010
Elgeyo-Marakwet	Trend only (A)	10.543	0.151	—	0.958	0.074
Embu	Trend + Elasticity (C)	11.312	0.091	0.033	0.992	0.021
Garissa	Trend only (A)	10.280	0.102	—	0.982	0.033
Homa Bay	Trend only (A)	11.041	0.104	—	0.991	0.023
Isiolo	Trend only (A)	9.474	0.103	—	0.989	0.025
Kajiado	Trend only (A)	11.255	0.097	—	0.990	0.023
Kakamega	Trend + Growth (D)	11.604	0.102	0.001	0.992	0.020
Kericho	Trend only (A)	11.364	0.096	—	0.986	0.028
Kiambu	Trend + Elasticity (C)	12.620	0.094	0.026	0.985	0.029
Kilifi	Trend + Growth (D)	11.634	0.089	0.000	0.991	0.016
Kirinyanga	Trend + Elasticity (C)	11.076	0.099	0.035	0.987	0.027
Kisii	Trend only (A)	11.503	0.105	—	0.977	0.038
Kisumu	Trend only (A)	11.793	0.094	—	0.994	0.019
Kitui	Trend + Elasticity (C)	11.073	0.109	0.074	0.950	0.063
Kwale	Trend + Elasticity (C)	10.926	0.112	0.114	0.997	0.012
Laikipia	Trend + Growth (D)	10.704	0.110	0.004	0.974	0.036
Lamu	Trend only (A)	9.702	0.103	—	0.987	0.032
Machakos	Trend only (A)	12.172	0.073	—	0.988	0.018
Makueni	Trend + Growth (D)	10.960	0.094	0.008	0.960	0.034
Mandera	Trend + Elasticity (C)	10.131	0.110	0.088	0.991	0.025
Marsabit	Trend only (A)	10.116	0.125	—	0.979	0.045
Meru	Trend only (A)	11.948	0.111	—	0.994	0.021
Migori	Trend only (A)	11.032	0.102	—	0.987	0.026
Mombasa	Trend + Elasticity (C)	12.538	0.081	0.152	0.989	0.020
Murang'a	Trend only (A)	11.533	0.099	—	0.994	0.019

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Nairobi	Trend only (A)	14.145	0.099	—	0.990	0.027
Nakuru	Trend only (A)	12.420	0.102	—	0.970	0.039
Nandi	Trend only (A)	11.270	0.098	—	0.978	0.037
Narok	Trend only (A)	11.267	0.113	—	0.987	0.029
Nyamira	Trend only (A)	10.871	0.115	—	0.983	0.034
Nyandarua	Trend + Elasticity (C)	10.967	0.140	0.409	0.977	0.050
Nyeri	Trend only (A)	11.555	0.098	—	0.985	0.030
Samburu	Trend only (A)	9.549	0.104	—	0.967	0.048
Siaya	Trend only (A)	10.809	0.112	—	0.992	0.024
Taita Taveta	Trend only (A)	10.402	0.095	—	0.973	0.037
Tana River	Trend only (A)	9.588	0.102	—	0.973	0.042
Tharaka Nithi	Trend + Growth (D)	10.278	0.111	0.028	0.996	0.016
Trans Nzoia	Trend only (A)	11.274	0.104	—	0.984	0.030
Turkana	Trend + Growth (D)	10.836	0.105	0.084	0.980	0.032
Uasin Gishu	Trend only (A)	11.723	0.097	—	0.991	0.018
Vihiga	Trend only (A)	10.552	0.111	—	0.996	0.018
Wajir	Trend only (A)	10.126	0.104	—	0.973	0.041
West Pokot	Trend only (A)	10.447	0.110	—	0.973	0.042

Table 3 forecasting framework

County	Model	a	b	γ	OSR_2025 Forecast	OSR_2026 Forecast
Baringo	Trend only	10.539	0.102	—	EXP(a+13b)	EXP(a+14b)
Bomet	Trend only	11.098	0.123	—	EXP(a+13b)	EXP(a+14b)
Bungoma	Trend + elasticity	11.555	0.100	0.054	$EXP(a+13b+\gamma \cdot lnGCP_{(2025)})$	$EXP(a+14b+\gamma \cdot lnGCP_{(2026)})$
Busia	Trend only	10.623	0.113	—	EXP(a+13b)	EXP(a+14b)
Elgeyo-Marakwet	Trend only	10.543	0.151	—	EXP(a+13b)	EXP(a+14b)
Embu	Trend + elasticity	11.312	0.091	0.033	$EXP(a+13b+\gamma \cdot lnGCP_{(2025)})$	$EXP(a+14b+\gamma \cdot lnGCP_{(2026)})$
Garissa	Trend only	10.280	0.102	—	EXP(a+13b)	EXP(a+14b)
Homa Bay	Trend only	11.041	0.104	—	EXP(a+13b)	EXP(a+14b)
Isiolo	Trend only	9.474	0.103	—	EXP(a+13b)	EXP(a+14b)
Kajiado	Trend only	11.255	0.097	—	EXP(a+13b)	EXP(a+14b)
Kakamega	Trend + growth	11.604	0.102	0.001	$EXP(a+13b+\gamma \cdot \Delta lnGCP_{(2025)})$	$EXP(a+14b+\gamma \cdot \Delta lnGCP_{(2026)})$
Kericho	Trend only	11.364	0.096	—	EXP(a+13b)	EXP(a+14b)
Kiambu	Trend + elasticity	12.620	0.094	0.026	$EXP(a+13b+\gamma \cdot lnGCP_{(2025)})$	$EXP(a+14b+\gamma \cdot lnGCP_{(2026)})$
Kilifi	Trend + growth	11.634	0.089	0.000	EXP(a+13b)	EXP(a+14b)
Kirinyanga	Trend + elasticity	11.076	0.099	0.035	$EXP(a+13b+\gamma \cdot lnGCP_{(2025)})$	$EXP(a+14b+\gamma \cdot lnGCP_{(2026)})$
Kisii	Trend only	11.503	0.105	—	EXP(a+13b)	EXP(a+14b)
Kisumu	Trend only	11.793	0.094	—	EXP(a+13b)	EXP(a+14b)
Kitui	Trend + elasticity	11.073	0.109	0.074	$EXP(a+13b+\gamma \cdot lnGCP_{(2025)})$	$EXP(a+14b+\gamma \cdot lnGCP_{(2026)})$

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Kwale	Trend + elasticity	10.926	0.112	0.114	EXP(a+13b+γ·lnGCP$_{(2025)}$)	EXP(a+14b+γ·lnGCP$_{(2026)}$)
Laikipia	Trend + growth	10.704	0.110	0.004	EXP(a+13b+γ·ΔlnGCP$_{(2025)}$)	EXP(a+14b+γ·ΔlnGCP$_{(2026)}$)
Lamu	Trend only	9.702	0.103	—	EXP(a+13b)	EXP(a+14b)
Machakos	Trend only	12.172	0.073	—	EXP(a+13b)	EXP(a+14b)
Makueni	Trend + growth	10.960	0.094	0.008	EXP(a+13b+γ·ΔlnGCP$_{(2025)}$)	EXP(a+14b+γ·ΔlnGCP$_{(2026)}$)
Mandera	Trend + elasticity	10.131	0.110	0.088	EXP(a+13b+γ·lnGCP$_{(2025)}$)	EXP(a+14b+γ·lnGCP$_{(2026)}$)
Marsabit	Trend only	10.116	0.125	—	EXP(a+13b)	EXP(a+14b)
Meru	Trend only	11.948	0.111	—	EXP(a+13b)	EXP(a+14b)
Migori	Trend only	11.032	0.102	—	EXP(a+13b)	EXP(a+14b)
Mombasa	Trend + elasticity	12.538	0.081	0.152	EXP(a+13b+γ·lnGCP$_{(2025)}$)	EXP(a+14b+γ·lnGCP$_{(2026)}$)
Murang'a	Trend only	11.533	0.099	—	EXP(a+13b)	EXP(a+14b)
Nairobi	Trend only	14.145	0.099	—	EXP(a+13b)	EXP(a+14b)
Nakuru	Trend only	12.420	0.102	—	EXP(a+13b)	EXP(a+14b)
Nandi	Trend only	11.270	0.098	—	EXP(a+13b)	EXP(a+14b)
Narok	Trend only	11.267	0.113	—	EXP(a+13b)	EXP(a+14b)
Nyamira	Trend only	10.871	0.115	—	EXP(a+13b)	EXP(a+14b)
Nyandarua	Trend + elasticity	10.967	0.140	0.409	EXP(a+13b+γ·lnGCP$_{(2025)}$)	EXP(a+14b+γ·lnGCP$_{(2026)}$)
Nyeri	Trend only	11.555	0.098	—	EXP(a+13b)	EXP(a+14b)
Samburu	Trend only	9.549	0.104	—	EXP(a+13b)	EXP(a+14b)
Siaya	Trend only	10.809	0.112	—	EXP(a+13b)	EXP(a+14b)
Taita Taveta	Trend only	10.402	0.095	—	EXP(a+13b)	EXP(a+14b)
Tana River	Trend only	9.588	0.102	—	EXP(a+13b)	EXP(a+14b)
Tharaka Nithi	Trend + growth	10.278	0.111	0.028	EXP(a+13b+γ·ΔlnGCP$_{(2025)}$)	EXP(a+14b+γ·ΔlnGCP$_{(2026)}$)
Trans Nzoia	Trend only	11.274	0.104	—	EXP(a+13b)	EXP(a+14b)
Turkana	Trend + growth	10.836	0.105	0.084	EXP(a+13b+γ·ΔlnGCP$_{(2025)}$)	EXP(a+14b+γ·ΔlnGCP$_{(2026)}$)
Uasin Gishu	Trend only	11.723	0.097	—	EXP(a+13b)	EXP(a+14b)
Vihiga	Trend only	10.552	0.111	—	EXP(a+13b)	EXP(a+14b)
Wajir	Trend only	10.126	0.104	—	EXP(a+13b)	EXP(a+14b)
West Pokot	Trend only	10.447	0.110	—	EXP(a+13b)	EXP(a+14b)

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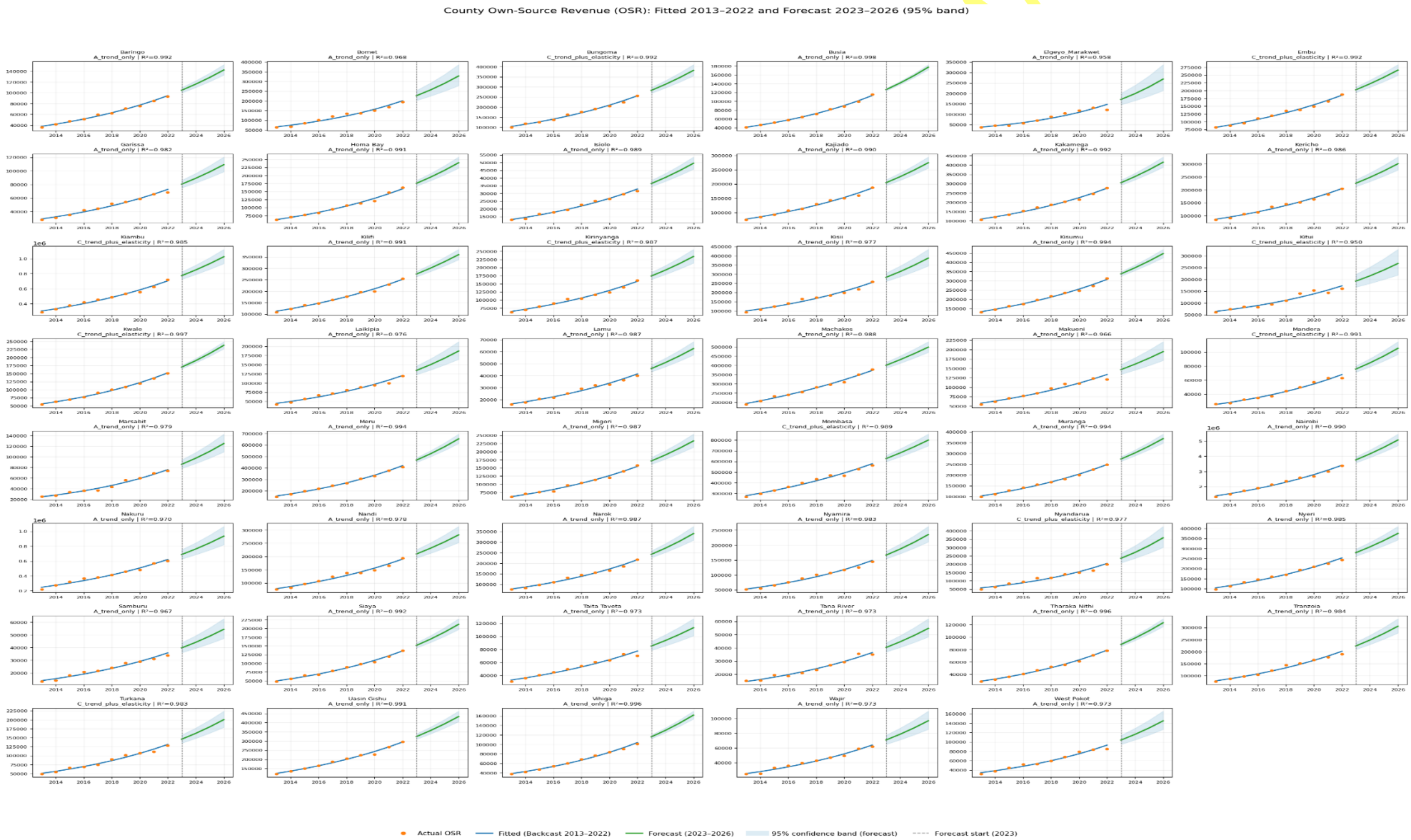
Figure 9 Back casting and forecasting visualization



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The fitted backcasts for 2013–2022 (figure 9) demonstrate a high degree of accuracy across counties, with the selected optimal models closely tracking observed Own-Source Revenue (OSR) outturns. In most counties, the fitted values align tightly with actual collections, reflected in consistently high coefficients of determination (R^2 typically above 0.95) and low in-sample errors. This indicates that historical OSR dynamics are well captured by simple log-linear trend specifications, with elasticity-augmented models applied selectively where economic scale effects are statistically relevant. Importantly, deviations between actual and fitted values are small and largely unsystematic, suggesting limited structural breaks over the estimation period. The narrow 95 percent confidence bands around the forecasts for 2023–2026 further indicate low estimation uncertainty, reinforcing confidence in the short- to medium-term projections. Overall, the strong in-sample performance provides a credible basis for forward-looking OSR forecasts, while the widening of confidence bands over the forecast horizon appropriately reflects increasing uncertainty over time rather than model instability

8 Revenue Structure by Tax Head and Implications for Forecasting

Figure 10 completes the analytical framework by examining OSR composition at the tax-head level. The stacked bars show that, despite year-to-year fluctuations in total collections, the structure of county revenue is remarkably stable over time. Hospital fees, single business permits, land rates, parking fees, and market-related charges consistently dominate OSR, with only marginal reallocation across heads between 2019/20 and 2023/24. This persistence indicates that counties operate within a relatively fixed institutional and legal revenue architecture, where changes in total OSR are driven more by scale and administration than by shifts in the underlying tax mix.

This structural stability has direct forecasting implications. Because the relative shares of tax heads change slowly, it is both practical and defensible to forecast total OSR first, using the optimal county-level models identified earlier, and then distribute projected totals across tax heads using the most recent observed structure (or a smoothed average of recent years). This approach aligns naturally with budgeting and Medium-Term Revenue Frameworks, where expenditure plans are already mapped to relatively stable revenue categories.

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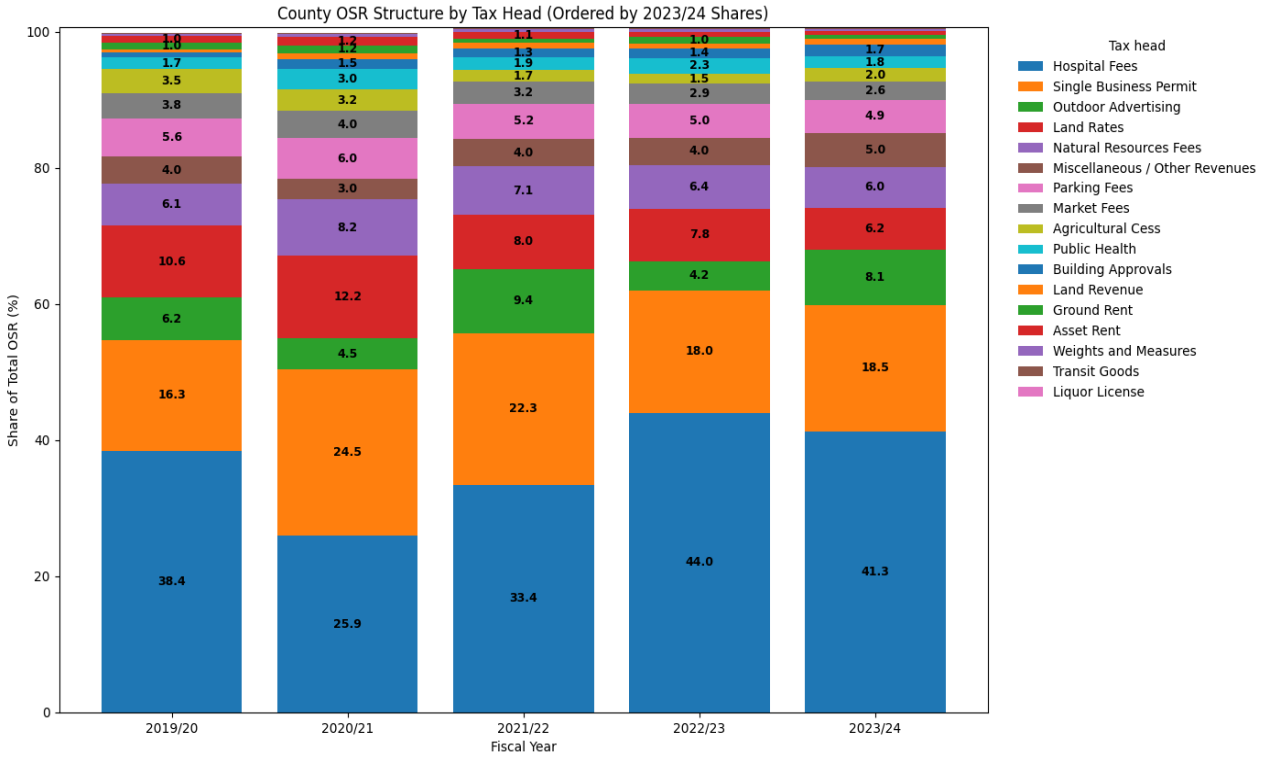
Figure 10 Tax structure



By contrast, attempting to link individual tax heads directly to GCP performs poorly. As shown in earlier sections, GCP–OSR correlations are weak and inconsistent at the aggregate level, and they become even noisier when disaggregated by tax head. Different heads respond to different drivers—administrative enforcement, service demand, policy changes, and fees—rather than to aggregate economic output. As a result, tax-head-level elasticities with respect to GCP are unstable, often sign-switching, and lack policy interpretability.

The conclusion is clear: point elasticities—whether applied to total OSR or to individual tax heads—are not suitable for county revenue forecasting. A structure-consistent approach performs better: forecast total OSR using empirically validated county models, then allocate revenues by tax head based on stable observed shares. This preserves realism, reduces noise, and produces forecasts that are credible for budgeting, appraisal, and intergovernmental fiscal planning.